Anti-Money Laundering (AML) & Transaction Monitoring Policy

Dataset Scope: IBM Transactions for AML (Small Dataset – ~7 Million Transactions)
Version: 1.0
Effective Date: 2026

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1. Purpose

This policy establishes automated controls to detect, flag, and classify suspicious financial activity using the following dataset columns:

- Timestamp
- From Bank
- Account (From Account – Hexadecimal)
- To Bank
- Account (To Account – Hexadecimal)
- Amount Received
- Receiving Currency
- Amount Paid
- Payment Currency
- Payment Format

The objective is to identify:

- Structuring (smurfing)
- High-value suspicious transfers
- Rapid movement of funds
- Cross-bank laundering chains
- Currency inconsistencies
- Suspicious transaction formats

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2. Data Integrity & Validation Rules

These rules ensure the dataset is clean before AML detection begins.

Rule 2.1 – Mandatory Field Policy

A violation is triggered if any of the following fields are NULL or empty:

- Timestamp
- From Bank
- Account (From Account)
- To Bank
- Account (To Account)
- Amount Paid
- Payment Currency

Violation Type: DATA_MISSING_CRITICAL

Rule 2.2 – Amount Consistency Rule

If:

- Amount Paid ≤ 0
  OR
- Amount Received ≤ 0

Trigger violation.

Violation Type: INVALID_TRANSACTION_AMOUNT

Rule 2.3 – Currency Mismatch Rule

If:

- Receiving Currency ≠ Payment Currency
  AND no exchange indicator exists

Flag as suspicious.

Violation Type: CURRENCY_MISMATCH

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3. Transaction Monitoring Rules

Rule 3.1 – High Value Transaction Rule

Based on your dataset distribution, suspicious high bands begin above:

- 150,000 currency units
- 200,000+ currency units
- 300,000+ currency units

If:

Amount Paid ≥ 150000

Flag as high risk.

If:

Amount Paid ≥ 300000

Flag as critical risk.

Violation Type: HIGH_VALUE_TRANSFER

Rule 3.2 – Structuring (Smurfing) Detection

If:

- Same Account (From Account)
- More than 5 transactions
- Within 24 hours
- Each Amount Paid between 7000 and 15000
- Total sum > 50000

Flag as structuring attempt.

Violation Type: STRUCTURING_ACTIVITY

Rule 3.3 – Rapid Fund Movement Rule

If:

- Account receives funds
- Same account sends funds within 1 hour
- Sent amount ≥ 90% of received amount

Flag as suspicious layering.

Violation Type: RAPID_LAYERING

Rule 3.4 – Same Bank Round Tripping

If:

From Bank = To Bank
AND From Account ≠ To Account
AND Amount Paid ≥ 100000

Flag as potential internal laundering.

Violation Type: ROUND_TRIP_INTERNAL

Rule 3.5 – Cross-Bank Large Transfer Rule

If:

From Bank ≠ To Bank
AND Amount Paid ≥ 200000

Flag as cross-border laundering risk.

Violation Type: CROSS_BANK_LARGE_TRANSFER

Rule 3.6 – Repeated Transfers to Same Destination

If:

Same From Account
Sending to same To Account
More than 10 times

Within 7 days
Total value ≥ 250000

Flag as suspicious funneling.

Violation Type: FUNNELING_PATTERN

Rule 3.7 – Suspicious Payment Format Rule

If Payment Format is:

- cheque
- manual
- cash

AND Amount Paid ≥ 100000

Flag as suspicious format usage.

Violation Type: SUSPICIOUS_PAYMENT_FORMAT

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4. Risk Classification Model

Each violation contributes to risk score:

Violation	Risk Score
DATA_MISSING_CRITICAL	30
INVALID_TRANSACTION_AMOUNT	40
CURRENCY_MISMATCH	50
HIGH_VALUE_TRANSFER	60
STRUCTURING_ACTIVITY	70
RAPID_LAYERING	80
ROUND_TRIP_INTERNAL	65
CROSS_BANK_LARGE_TRANSFER	75

Violation	Risk Score
FUNNELING_PATTERN	85
SUSPICIOUS_PAYMENT_FORMAT	55

Risk Levels

- 0–49 → Low Risk
- 50–99 → Medium Risk
- 100–149 → High Risk
- 150+ → Critical Risk

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5. Alert Generation Framework

When violation occurs, system must log:

- Timestamp
- From Bank
- Account (From Account)
- To Bank
- Account (To Account)
- Amount Paid
- Violation Type
- Risk Score

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6. Escalation Policy

Risk Level	Action
Medium	Manual Review
High	Compliance Review

Risk Level	Action
Critical	Immediate SAR (Suspicious Activity Report)

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7. Implementation Notes for Your Engine

For your backend:

- Group by Account (From Account) for structuring
- Use Timestamp parsing for hourly/daily window checks
- Use aggregation pipeline for sum and count
- Maintain rolling 24-hour and 7-day windows
- Store violations in a separate collection like:
    - transaction_violations

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8. Example Violation Logic (Conceptual)

If:

Amount Paid = 320000
From Bank ≠ To Bank

Violations triggered:

- HIGH_VALUE_TRANSFER
- CROSS_BANK_LARGE_TRANSFER

Total Risk Score = 60 + 75 = 135
→ High Risk